

THUY LINH DO

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EDUCATION

- **Tulane University** Jul 2019 - May 2025
Doctor of Philosophy - PhD in Mathematics New Orleans, LA, USA
 - GPA: 4.00/4.00
- **Vietnam National University - University of Education** Sep 2013 - Jul 2017
Bachelor in Mathematics Teacher Education, Valedictorian Hanoi, Vietnam
 - GPA: 3.81/4.00

EMPLOYMENT HISTORY

- **Tulane University** 2025 - present
Research Scientist (remote) New Orleans, LA, USA
- **Tulane University** 2019 - 2025
PhD Candidate/Graduate Teaching Associate/Graduate Research Assistant New Orleans, LA, USA
- **NSF-Simons Southeast Center for Mathematics and Biology** 2022 - 2025
Junior Researcher Atlanta, GA, USA
- **Vietnam National University, University of Science** 2017 - 2019
Lecturer Hanoi, Vietnam

RESEARCH INTERESTS

Bayesian Hierarchical Models, Empirical Bayes, Change-Point Analysis, Intracellular Transport, Machine Learning.

TEACHING TRAINING AND EXPERIENCE

- **Graduate Teaching Assistant** 2019 - 2025
Tulane University - Department of Mathematics
 - **MATH 1110: Probability and Statistics** - conducted tutorial sessions to reinforce lecture content; created and graded weekly quizzes; graded homework, midterms, and final exams; held office hours and answered student questions.
 - **MATH 1210: Calculus I** - led problem-solving sessions and review workshops; graded homework and exams; held office hours.
 - **MATH 1220: Calculus II** - supported students in mastering advanced integration techniques; assisted in exam proctoring and grading; conducted tutorial sessions.
 - **MATH 2210: Calculus III** - facilitated group discussions and practice sessions; graded assignments and exams; provided individualized academic support during office hours.
 - **MATH 1230: Statistics for Scientists (Instructor)** - designed and delivered lectures tailored for science majors; developed quizzes, midterms, and final exams; assessed student performance and provided constructive feedback; integrated real-world scientific data into statistical examples; oversaw classroom management and student engagement.
- **Lecturer** 2017-2019
Vietnam National University, University of Science
 - Applied Statistics
 - Mathematical Statistics and Probability
 - Calculus I, II, III
 - Linear Algebra
- **Part-time Teacher** 2017-2019
High School for Gifted Students, VNU University of Science

- Pre-Calculus
- Pre-Algebra
- Probability and Statistics

• B.S in Education with a six-month internship

2016-2017

High School of Education and Sciences, VNU University of Education

- Calculus
- Analysis
- Probability and Statistics

SKILLS

- **Programming:** Python, R, Maple, SQL, MatLab.
- **Python:** Fluent in using JAX, tskit, Numpy, Matplotlib, PyTorch.
- **R:** Fluent in using RStudio, including R Script, R Markdown, packages ggplot2, tidyverse, dplyr, stats, MASS, caret, tidymodels, parallel.
- **Data & Analysis Tools:** ImageJ, Fiji (video and image preprocessing), LaTeX, Microsoft Office.
- **Languages:** Strong reading, writing and speaking competencies for English and Vietnamese.

RESEARCH PUBLICATIONS

I= IN PREPARATION, B= BOOKS AND CHAPTERS, J=JOURNAL, S=IN SUBMISSION, T=THESIS

- [T] **Linh Do** (2025). **Changepoint detection: theory, algorithms, and application in intracellular transport**. Tulane University.
- [J.1] **Dat Do, Linh Do, XuanLong Nguyen** (2025). **Strong identifiability and parameter learning in regression with heterogeneous response**. *Electronic Journal of Statistics*, 19 (1) 131 - 203.
- [J.2] **Keisha J. Cook, Nathan Rayens, Linh Do, Christine K. Payne, and Scott A. McKinley** (2025). **Considering experimental frame rates and robust segmentation analysis of piecewise-linear microparticle trajectories**. *Mathematical Biosciences and Engineering*, Volume 22, Issue 10: 2595-2626..
- [I.1] **Linh Do, Dat Do, Keisha J. Cook, Nathan Rayens, Christine K. Payne, Scott A. McKinley** (2025). **Change-in-velocity detection for multidimensional data**. . To be submitted to *Journal of Computational and Graphical Statistics*.  code
- [S.1] **Dat Do, Linh Do, Scott A. McKinley, XuanLong Nguyen, Jonathan Terhorst** (2024). **Dendrogram of mixing measures: Hierarchical Clustering and model selection for finite mixture models**. The manuscript is under submission to *Biometrika*.  R package
- [B.1] **Nguyen, A.T., Pham, H.T.T, Trinh, Q.A, Linh Do, Dang, P.A, & Hens, L.,** (2021). **The geography of climate change adaptation in the vietnam northern mountains: A quantitative analysis for intentions of indigenous ethnic minorities using structural equation modeling (sem) and protection motivation theory (pmt)**. In: Rai, P.K., Singh, P., Mishra, V.N. (eds) *Recent Technologies for Disaster Management and Risk Reduction*. Earth and Environmental Sciences Library. Springer, Cham.
- [I.2] **Linh Do and et al.** **Dendrogram of signal functions for summarization and model selection in changepoint detection**. In preparation for submission to the *Journal of Statistical Planning and Inference*.
- [I.3] **Dat Do, Linh Do, Peter Carbonetto, Matthew Stephens.** **Improved out-of-sample covariance estimation for fitting the SuSiE model with summary statistics**. In preparation.
- [I.4] **Dat Do, Linh Do, Sunrit Chakraborty.** **Visualizing, Interpreting, and Improving Sparse Matrix Factorization Model Fit**. In preparation.

PRESENTATIONS

- **A segmentation analysis in noisy trajectories**. NITMB MathBio Convergence Conference. *Aug 2025*.
- **CPLASS: A segmentation analysis in noisy trajectories**. Final Symposium, NSF-Simons Southeast Center for Mathematics and Biology. *Apr 2025*.
- **Inferring and interpreting heterogeneous data using dendrograms**. Computational and Applied Math Seminar at the Department of Mathematics, Tulane University. *Nov 2024*.
- **Dendrogram of mixing measures: Hierarchical clustering and model selection for finite mixture models**. Joint Statistical Meetings (Portland, Oregon). *Aug 2024*
- **A new changepoint algorithm detecting multiple change-in-velocity**. NSF-Simons Southeast Center for Mathematics and Biology (SCMB). *Oct 2023*.
- **Mixture Model and the applications**. Graduate Colloquium at the Department of Mathematics, Tulane University. *Nov 2022*.
- **Strong identifiability and parameter learning in regression with heterogeneous response**. 13th International Conference on Bayesian Nonparametrics (Puerto Varas, Chile). *Oct 2022*.

HONORS AND AWARDS

- **Mathematical Sciences and Research Institute (MSRI) Summer Program Admission** 2022
MSRI & Courant Institute in New York
 - The award was being selected into the program. Mathematics of Machine Learning (INdAM and Courant Institute).
- **Toshiba Scholarship for young scientists** 2018
Toshiba
 - The Toshiba scholarship program aims to support students, graduate students, and graduate students (students) with excellent academic and research achievements.
- **Golden record of excellent valedictorians graduating from universities in Hanoi, Vietnam** 2017
Hanoi Municipal Party Committee - Hanoi People's Council - Hanoi People's Committee.
 - The program to honor outstanding valedictorians graduating from universities and academies in Hanoi in 2017, organized by the Hanoi Municipal Party Committee, the People's Council, and the People's Committee.
- **Prize for Outstanding Student Performance** 2017
Hanoi University of Science and Hanoi University of Education, VNU
 - For students with excellent achievement in learning and scientific research.

VOLUNTEER EXPERIENCE

- **Session Leader** 2020
Boys at Tulane in STEM (BATS) - K-12 STEM Education Events - Tulane University - New Orleans - Louisiana.
- **Conference Organizer, Marketing and Photography** 2019
Math For All - Tulane University - New Orleans - Louisiana.

REFERENCES

1. **Scott A. McKinley**
Associate Professor, Department of Mathematics, Tulane University
Email: scott.mckinley@tulane.edu
Phone: +1-504-862-3246
Role: Thesis Advisor / Chair of Dissertation Committee
2. **XuanLong Nguyen**
Professor, Department of Statistics, University of Michigan
Email: xuanlong@umich.edu
Phone: +1-734-763-3499
Role: Professor
3. **Michael Joyce**
Senior Professor of Practice, Department of Mathematics, Tulane University
Email: mjoyce3@tulane.edu
Phone: +1-504-862-3438
Role: Teaching Advisor/ Professor
4. **Lifeng Han**
Professor of Practice, Department of Mathematics, Tulane University
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Role: Professor
5. **Viet Cuong Do**
Associate Professor, Department of Mathematics, Vietnam National University, Hanoi University of Science
Email: vcuong.do@gmail.com
Phone: +84 868961764
Role: Professor